

6. [Maximum mark: 6]

A linear system of equations is given by

$$x - z = 4$$

$$2x + 2y + z = 14$$

$$x + 2y + \alpha z = \beta$$

Find the value of α and the value of β , where $\alpha, \beta \in \mathbb{Z}$, for which the system has an infinite number of solutions.

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